Karan Uppal

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Education

Indian Institute of Technology (IIT) Kharagpur, India — **CGPA: 9.16/10.0**
Major: Combined BSc. & MSc. in Mathematics and Computing — 2019 – 2024

Selected Publications

*denotes equal contribution

MoVE: Mixture-of-Vocabulary-Experts for Improved Representation Learning — *Under Review (ICLR)*
***Karan Uppal**, Nagarajan Natarajan, Manik Varma*

Understanding Task Transfer in Vision-Language Models — *Under Review (CVPR)*
Bhuvan Sachdeva, **Karan Uppal***, Abhinav Java*, Vineeth N. B.* | NeurIPS 2025 UniReps Workshop

Decoding Attention from Gaze: A Benchmark Dataset and End-to-End Models
***Karan Uppal**, Jaeah Kim, Shashank Singh* | NeurIPS 2022 Gaze Meets ML Workshop

[RE] Background-Aware Pooling & Noise-Aware Loss for Weakly-Supervised Semantic Segmentation
Aryan Mehta, **Karan Uppal*** et al.* | ReScience C Journal 2021 | Journal Showcase NeurIPS 2022

Research Experience

Research Fellow | Microsoft Research India — **Bengaluru, India**
Guide: Dr. Nagarajan Natarajan, Prof. Vineeth N Balasubramanian, Prof. Manik Varma — Jul 2024 — Present

- Scaled vocabulary sizes beyond 1M using MoEs to pretrain dense retrieval models for Bing Ads
- Investigated task transferability in VLMs to uncover cross-task interactions and scaling effects

DAAD WISE Scholar | Max Planck Institute for Intelligent Systems — **Tuebingen, Germany**
Guide: Dr. Shashank Singh, Prof. Bernhard Schölkopf — May 2022 — Jul 2022

- Released a new dataset and baseline models to decode human attention from gaze data in road scenes
- Designed a novel CNN architecture to incorporate temporal gaze data for object localisation

Research Intern | Harvard University — **Cambridge, USA**
Guide: Prof. Hanspeter Pfister — May 2021 — Aug 2021

- Experimented with different variants of the 3D UNet model for instance segmentation on brain volumes
- Achieved 1st position on the CREMI leaderboard for synaptic cleft detection task using self-learning

Work Experience

AI Resident | Ema Unlimited — **Bengaluru, India**
Early stage startup for Agentic AI-powered employees — Jan 2024 — Apr 2024

- Created an agentic pipeline to convert natural language to SQL queries over enterprise databases
- Developed multi-LLM RAG-based ensemble framework for improved inference and accuracy

Summer Fellow | University of Warwick and UN-REDD — **Coventry, UK**
Data Science for Social Good (DSSG) — Jun 2023 — Aug 2023

- Implemented a spatiotemporal UNet for forecasting deforestation in the Brazilian Amazon
- Deployed a low-latency visualisation tool with feature ablation to analyze drivers of deforestation

Applied Research Intern | NVIDIA — **Bengaluru, India**
WWFO - Solution Architect and Engineering — Aug 2021 — Jan 2022

- Developed and deployed 5 scalable pipelines for the tasks of image classification and object detection
- Achieved 10x speed-up in model inference for QnA, Re-ranking & DPR for end-to-end applications

Technical Skills

Programming languages: C/C++, Python, Bash, SQL
ML/AI: PyTorch (Lightning), Transformers, Hydra
Web Technologies: Streamlit, GEE, JavaScript
Miscellaneous: Docker, Git, LaTeX, Markdown

Competitions and Challenges

Campus Super Bowl: Data Science Challenge — **American Express**
One of India's largest Data Science Competitions — Sep 2022 — Nov 2022

- Achieved 2nd place out of 3600+ teams for the task of predicting credit card customer default
- Attained 82.4% accuracy using a gradient boosting ensemble, battling dataset imbalance

[RE] Background-Aware Pooling & Noise-Aware Loss for Weak Supervision — **Papers with Code**
Machine Learning Reproducibility Challenge — Nov 2021 — Feb 2022

- Reviewed above CVPR 2021 publication for reproducibility of its claims; presented at NeurIPS '22
- Implemented Noise-Aware Loss from scratch achieving state-of-the-art results on PASCAL VOC 2012

Navigation and Planning in Unknown Indoor Environments — **IROS-RSJ**
Navigation and Manipulation Challenge — Jul 2021 — Sep 2021

- Attained 1st place for the task of navigation and manipulation in unknown environments
- Developed a probabilistic distance minimization algorithm for path planning; published at ICCR 2022

Achievements

TSG, IIT KGP	Awarded Honorable Mention for distinction in technology endeavours in campus	**2024**
DSSG Fellow	Selected internationally to pursue a project with the UN-REDD in Coventry, UK	**2023**
DAAD WISE	Awarded a €3000+ grant to pursue research in Germany as a visiting researcher	**2022**
INSPIRE Scholar	Awarded 60,000 INR by the Government of India for excellence in sciences	**2021**
JEE Main	Attained a percentile of 99.87% among more than 1.5 million applicants	**2019**
JEE Advanced	Ranked in the top 0.82% among the 200k shortlisted candidates from above	**2019**

Relevant Coursework

* indicates MOOCs

Artificial Intelligence: AI Foundations, Machine Learning Foundations*, Deep Learning Foundations, Genetic Algorithms, Computer Vision*, Graphical and Generative Models, Dependable and Secure AI-ML

Computer Science: Design and Analysis of Algorithms, Object-Oriented Systems Design, Computer Architecture, Systems Programming, Operating Systems, File Organization and Database Systems

Mathematics: Probability and Statistics, Linear Algebra, Stochastic Processes, Real Analysis, Functional Analysis, Measure Theory and Integration, Modern Algebra, Optimization Techniques, Graph Theory

Positions of Responsibility

Volunteership — **IIT Kharagpur**
Independent Endeavours — Aug 2022 — Nov 2022

- Co-organised a donation drive for an Orphanage in IIT Kharagpur, raising over 55,000 INR
- Conducted a blood donation drive with 150+ participants across the campus

Deep Learning Team Lead — **IIT Kharagpur**
Autonomous Ground Vehicle Research Group — Mar 2021 — May 2022

- Headed a team of 6 sophomores for participation in Machine Learning Reproducibility Challenge 2021
- Organised over 10 sessions for a team of 15 freshers, for training in Image Processing and Deep Learning

Head Analytics — **IIT Kharagpur**
Business Club — Jul 2021 — May 2022

- Organised the 4th international edition of our flagship event, witnessing 3000+ participation
- Conducted frequent knowledge sessions for a team of 30+ members on the topics of Machine Learning

Extracurriculars

Open Source : Contributed to the huggingface/transformers repository for bug fixes in MoE architecture
Reproducible Science : Authored manuscript for ML Reproducibility Challenge, presented at NeurIPS 2022
Technical Writing : Writer of a blog series summarizing research papers in the field of deep learning
Teaching : Mentored children of economically weaker section of 6th to 8th standards in academics